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Statistics

Central tendency: the descriptive summary of a dataset
Through a single value from the dataset, it reflects the centre of distribution.
It doesn't provide information regarding individual data from the dataset.

Mean: average of the given #s and is calculated by dividing the sum of the #s divided by the total # of #s.

e.g. mean of 1, 2, 4, 8, 10, $\bar{x} = \frac{\sum x}{n} = \frac{25}{5} = 5$

mean for ungrouped data = $\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$

mean for grouped data = $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$

e.g.

x_i	11	14	17	20
f_i	3	6	8	7

$\bar{x} = \frac{11 \cdot 3 + 14 \cdot 6 + 17 \cdot 8 + 20 \cdot 7}{3+6+8+7} = \frac{393}{24} = 16.375$

e.g.

Data range	frequency	x_i
1-5	3	3
6-10	2	8
11-15	2	13
16-20	4	18

$\bar{x} = \sum f_i x_i$
$= \frac{3 \cdot 3 + 8 \cdot 2 + 13 \cdot 2 + 18 \cdot 4}{3+2+2+4} = \frac{123}{11} = 11.1818$

Geometric Mean: $GM = \sqrt[n]{x_1 \, x_2 \, x_3 \cdots x_n}$

e.g. GM of 2, 3, 4 = $\sqrt[3]{2 \cdot 3 \cdot 4} = 2\sqrt[3]{3}$

Harmonic Mean:
used to average ratios $H.M. = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}$

Root Mean Square (quadratic mean): $x_{rms} = \sqrt{\frac{x_1^2 + x_2^2 + \cdots + x_n^2}{n}}$

Q. In her last basketball game, Jackie scored 36 points.
This raised her mean points per game from 20 to 21.
To raise this average to 22 points, how many more points must she score in the next game?

→ $\underbrace{x_1, x_2, \cdots, x_n}_{\bar{x}=20}, 36, y$

$\bar{x} = 21$

$\bar{x} = 22$

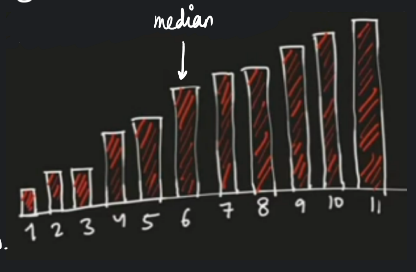
$\sum_{i=1}^{n} x_i = 20n$

$\sum x_i + 36 = 21(n+1)$
$20n + 36 = 21n + 21$
$n = 15$
$\sum_{i=1}^{n} x_i = 20 \cdot 15 = 300$

$22 = \frac{300 + 36 + y}{17}$

$y = 38$

Median:
It is the middlemost or centre value in a set. It is the # that is halfway into the set.
To find the median, the data should be arranged in ascending order.



For ungrouped data:

- odd # of observations, median = $\left(\frac{n+1}{2}\right)^{th}$ term

- even # of observations, median = avg. of $\frac{n}{2}^{th}$ term and $\left(\frac{n}{2}+1\right)^{th}$ term.

e.g. median of 63, 55, 14:
sorted: 14, 55, 63; median = 55

median of 1, 3, 5, 7, 9, 10 = avg(3rd and 4th items after sorting) = avg(5, 7) = 6

For grouped data:

- discrete:

e.g. the # of goals scored by a player during a hockey season:

# of goals	frequency
0	1
1	6
2	7
3	2
4	3
5	1

→

# of goals	frequency	cum freq
0	1	1
1	6	7
2	7	14 ← those would lie here
3	2	16
4	3	19
5	1	20

20 data points. ∴ we need 10th and 11th points.

median = avg (2, 2) = 2

For grouped data:

- ranged/continuous:

e.g. the marks obtained by 17 students are recorded. Median?

marks	freq
0-10	2
10-20	4
20-30	5
30-40	4
40-50	2

→

marks	freq	cumfreq	
0-10	2	2	
10-20	4	6	
20-30	5	11	← median class
30-40	4	15	
40-50	2	17	

median = $\left(\frac{17}{2}\right)^{th}$ = 8.5th element

in the grouped data given, it isn't possible to locate the middlemost observation by simple

Median class:
The class whose cumulative frequency

$$\text{Median} = l + \frac{\frac{N}{2} - F}{f} \times h$$

l = lower limit of median class
f = freq. of median class
F = cumulative freq. of class preceding the median class
N = total # of observations
h = width of median class

Q. Find median of first 50 natural #s.

→ even # of observations.
median = avg ($\frac{N}{2}^{th}$ and $\frac{N}{2}+1^{th}$ data pts) = $\frac{25+26}{2} = 25.5$

Q. If 10, 13, 15, 18, x+1, x+3, 30, 32, 35, 41 are in ascending order with median 24, find x.

→ $\frac{x+1+x+3}{2} = 24 \Rightarrow x+2 = 24$
$x = 22$

eg: Find $\frac{k}{6}$ where 6 is the median of $\frac{k}{2}, \frac{k}{3}, \frac{k}{4}, \frac{k}{5}, \frac{k}{6}$

→ $\frac{k}{4} = 6 \Rightarrow \frac{k}{4} = 6$

Q. 100 students, each sowed 5 seeds into each of 100 planters. One week later, the # of seeds germinating in each planter was recorded.

# of seeds germinating	# of planters
0	10
1	20
2	30
3	25
4	10
5	5

i. What is the total # of seeds sown? → 500

ii. What fraction of seeds did not germinate?
→ # of germinated seeds = $0 \cdot 10 + 1 \cdot 20 + 2 \cdot 30 + 3 \cdot 25 + 4 \cdot 10 + 5 \cdot 5 = 220$
∴ # of seeds not germinated = $500 - 220 = 280$
∴ fraction of seeds not germinated = $\frac{280}{500} = 0.56 = 56\%$

iii. What is the mean # of seeds germinating in the planter?

→ $\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{220}{100} = 2.20$

iv. What is the median of the observations?

→

x_i	f_i	cum. freq
0	10	10
1	20	30
2	30	60 ←
3	25	85
4	10	95
5	5	100

$\text{median} = \frac{50^{th}\text{ term} + 51^{st}\text{ term}}{2}$

$= \frac{2+2}{2} = 2$

Mode:
The value that appears most frequently.
When there are 2 modes, the data set is called bimodal. lly, there's trimodal and multimodal.

For ungrouped data,
the value occuring the most is the mode.
If there are multiple values with the max frequency, all of them are the mode.

Match No.	1	2	3	4	5	6	7	8	9	10
# wickets	2	1	1	3	2	3	2	2	4	1

→ mode = 2

For grouped data,

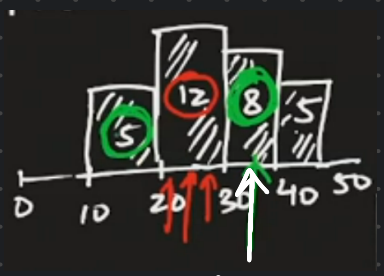
x_i	f_i
3	10
2	20
5	25
6	15

→ mode = 5

For grouped data (ranged),

Marks obtained	# students
10-20	5
20-30	12
30-40	8
40-50	5

← modal class 20 - 30



Both 20-30, the mode would lean towards the right.

$$\text{mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

f_1 : freq. of modal class

f_0 : freq. of preceding class

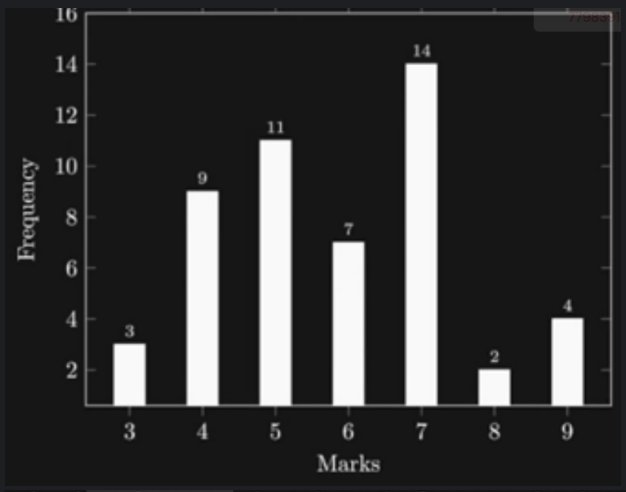
f_2 : freq. of succeeding class

for ques^n, $\text{mode} = 20 + \frac{12 - 5}{2 \cdot 12 - 5 - 8} \times 10$

$= 20 + \frac{70}{11} = 26.3636$

Empirical relationship betn mean, mode, and median:
In case of moderately skewed distribution, the difference between mean and mode is almost equal to 3 times the difference between mean and median.

mean - mode = 3 * (mean - median)



The above frequency chart shows the frequency distribution of marks obtained by a set of students in an exam.

From the data presented above, which one of the following is CORRECT?

A. mean > mode > median
B. mode > median> mean
C. mode > mean > median
D. median > mode > mean

→ mode = 7

mean: $\frac{3\cdot3+4\cdot9+5\cdot11+6\cdot7+7\cdot14+8\cdot2+9\cdot4}{3+9+11+7+14+2+4} = \frac{292}{50} = 5.84$

median: $\frac{50}{2}$th = 25th value = 6.

BTW here the data doesn't look too skewed.
could've used the empirical formula. (not a good idea)

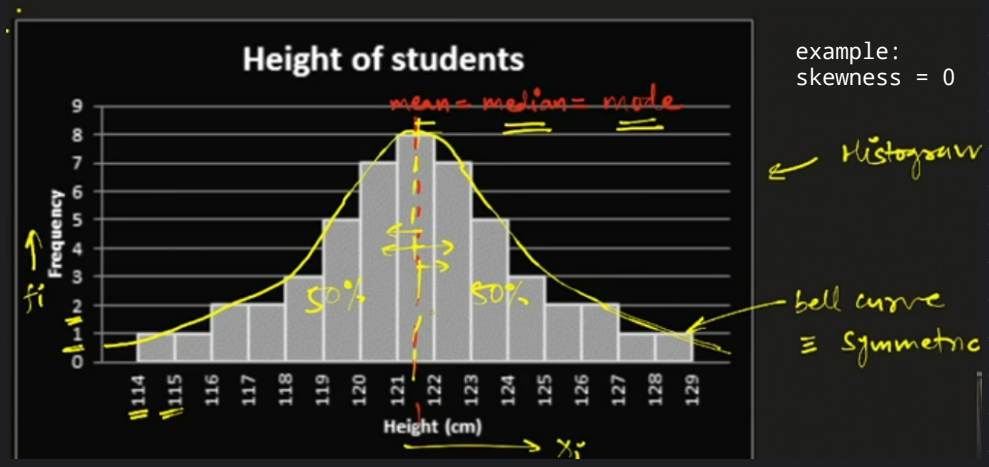
Measures of shape:

describe the distribution of the data within a dataset;
The distribution shape of quantitative data can be described as there is a logical order to the values and the low and high end values on the x-axis of the histogram.

Shape of a dataset:
A distribution of data item values may be symmetric or asymmetric.

Skewness

It is used as a measure of the asymmetry of the distribution.

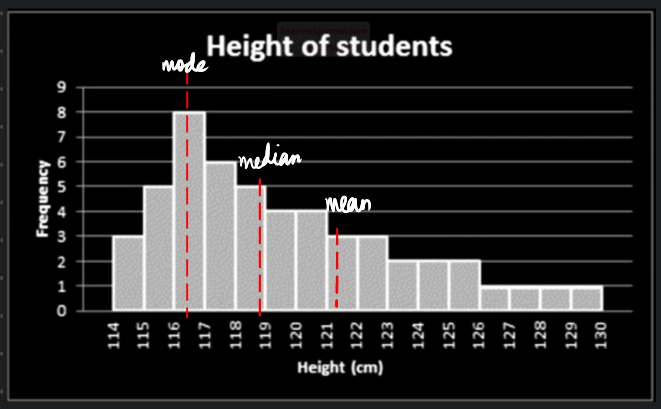


Symmetric:
In a symmetric distribution, the 2 sides of the mode of the distribution are the mirror images of one another.
- shape is symmetric
- mode = median = mean
- only one mode

Asymmetric:
In a symmetric distribution, the 2 sides of the mode of the distribution are not the mirror images of one another.

Skewness is the tendency for the values to be more frequent around the high and the low ends of the X-axis.

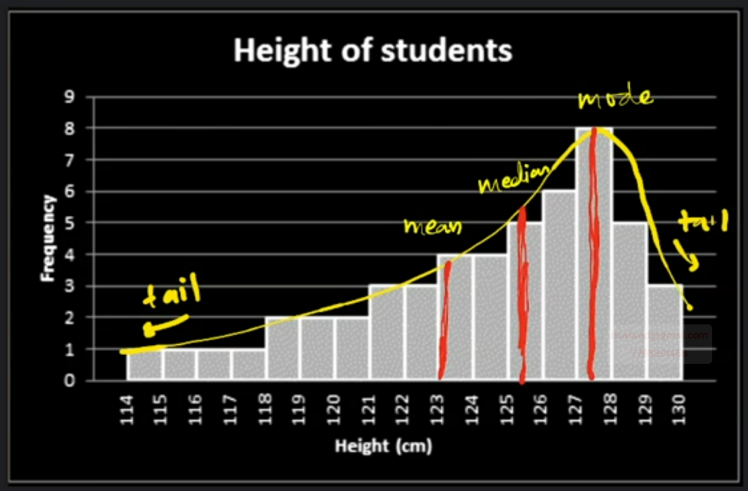
Positively skewed:
A dataset is said to be positively skewed when the tail on the right side is longer than that on the left.
Most of the values will be cluttered towards the left side of the X-axis, with increasingly fewer values on the right.



mean - mode = 3 * (mean - median)

For a +vely skewed dataset,
mean > mode > median

Negatively skewed:
A dataset is said to be negatively skewed when the tail on the left side of the histogram is longer than that on the right.
Most of the values are cluttered towards the right side of the X-axis, with increasingly fewer values on the left.



For a -vely skewed dataset,
mean < median < mode

$$\text{Pearson's 1st coefficient of skewness} = \frac{\text{mean} - \text{mode}}{\text{std. deviation}}$$

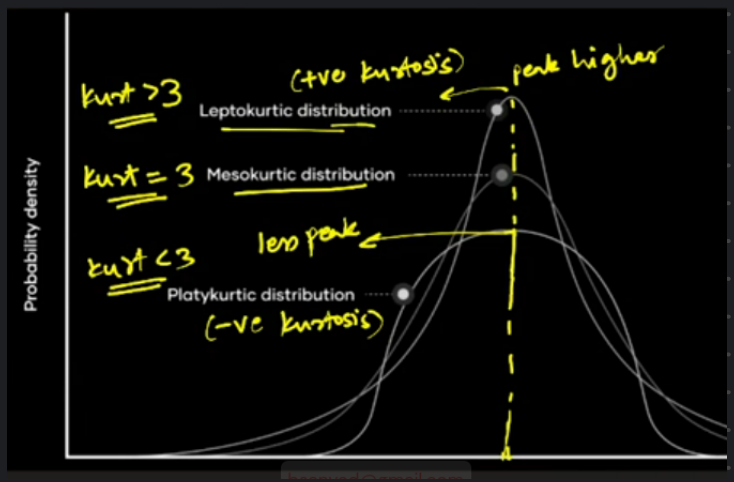
$$\text{Pearson's 2nd coefficient of skewness} = \frac{3(\text{mean} - \text{mode})}{\text{std. deviation}}$$

For symmetric data, skewness = 0.
If -0.5 <= skewness <= 0.5, the data is almost symmetric.
If -1 < skewness < -0.5 or 0.5 < skewness < 1, the data is moderately symmetric.
If skewness < -1 or skewness > 1, the data is highly symmetric.

Kurtosis

It is a statistical measure that quantifies the shape of a probability distribution.
It provides information about the tails and the peakedness of the distribution.

Positive kurtosis: heavier tails and more peaked distribution
Negative kurtosis: lighter tails and a flatter distribution



Peakedness in a data distribution is the degree to which the data values are concentrated around the mean.

Measures of spread

These describe how similar of varied the set of observations are for a particular value.

Example:

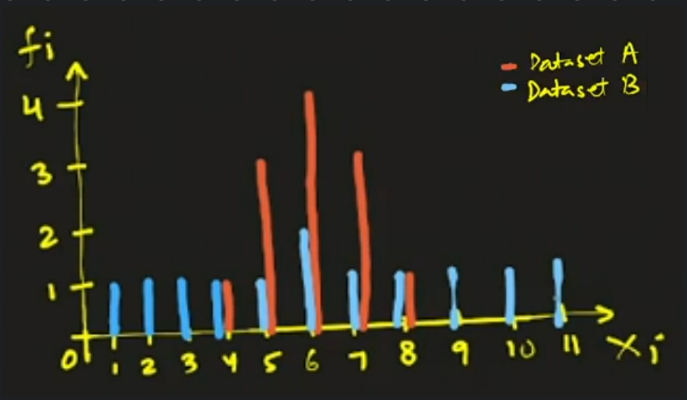
dataset A
4, 5, 5, 5, 6, 6, 6, 6, 7, 7, 7, 8

mean = 6,
median = 6,
mode = 6

dataset B
1, 2, 3, 4, 5, 6, 6, 7, 8, 9, 10, 11

mean = 6,
median = 6,
mode = 6

Both these datasets are the same in terms of measures of central tendency.
Both are symmetric as well.



The spread of the values in the graph tells us that the dataset B is more dispersed than A.

Range: difference between the largest and the smallest value in the dataset.
Range = largest value - smallest value

For dataset A, range = 8-4 = 4
For dataset B, range = 11-1 = 10
B is spread over 10 values, as opposed to A over 4 values.

Quartiles

These are 3 values that divide the # of data points into four parts, or four quarters, of more-or-less equal size.

In statistics, **quartiles** are a type of quantiles which divide the number of data points into four parts, or *quarters*, of more-or-less equal size. The data must be ordered from smallest to largest to compute quartiles; as such, quartiles are a form of order statistic. The three quartiles, resulting in four data divisions, are as follows:

- The first quartile (Q_1) is defined as the 25th percentile where lowest 25% data is below this point. It is also known as the *lower* quartile.
- The second quartile (Q_2) is the median of a data set; thus 50% of the data lies below this point.
- The third quartile (Q_3) is the 75th percentile where lowest 75% data is below this point. It is known as the *upper* quartile, as 75% of the data lies below this point.[1]

E.g. for the data points
2, 2, 4, 5, 5, 5, 8, 9, 9, 9, 12

Q_1 (median of left-half) Q_2 (median) Q_3

E.g. for the data points
1, 2, 3, 3, 5, 6, 6, 7

$Q_1 = \frac{2+3}{2} = 2.5$ $\quad Q_2 = \frac{3+5}{2} = 4$ $\quad Q_3 = \frac{6+6}{2} = 6$

For the example datasets A and B:

dataset A
4, 5, 5, 5, 6, 6, 6, 6, 7, 7, 7, 8
$Q_1 = 5$ $\quad Q_2 = 6$ $\quad Q_3 = 7$

dataset B
1, 2, 3, 4, 5, 6, 6, 7, 8, 9, 10, 11
$Q_1 = 3.5$ $\quad Q_2 = 6$ $\quad Q_3 = 8.5$

Interquartile range:

It is the difference between the upper (Q3) and the lower (Q1) quartiles.
It is considered a better measure of spread than the range as it is not affected by outliers.

dataset A
4, 5, 5, 5, 6, 6, 6, 6, 7, 7, 7, 8
$Q_1 = 5$ $\quad Q_2 = 6$ $\quad Q_3 = 7$

$IQR = Q_3 - Q_1 = 2$

dataset B
1, 2, 3, 4, 5, 6, 6, 7, 8, 9, 10, 11
$Q_1 = 3.5$ $\quad Q_2 = 6$ $\quad Q_3 = 8.5$

$IQR = Q_3 - Q_1 = 5$

Variance and standard deviation:

These are measures of spread of data around the mean. They summarize how close each observed data value is to the mean.
In datasets with a small spread, all values are close to the mean, resulting in a low variance.

The population variance of a discrete set of #s is:

$$\sigma^2 = \sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N}$$

$$\text{std dev}^n: \quad \sigma = \sqrt{\sum_{i=1}^{N} \frac{(x_i - \mu)^2}{N}}$$

x_i: i^{th} unit of the dataset
μ: population mean
N: # of observations

dataset A
4, 5, 5, 5, 6, 6, 6, 6, 7, 7, 7, 8
$\mu = 6$

$$\sigma^2 = \frac{(4-6)^2 + (5-6)^2 + \dots + (8-6)^2}{12}$$

$$= \frac{2^2 + 3 \cdot 1^2 + 4 \cdot 0^2 + 3 \cdot 1^2 + 2^2}{12}$$

$$= \frac{8 + 6 + 0}{12} = \frac{14}{12} = 1.17$$

dataset B
1, 2, 3, 4, 5, 6, 6, 7, 8, 9, 10, 11
$\mu = 6$

$$\sigma^2 = \frac{5^2 + 4^2 + 3^2 + 2^2 + 1^2 + 0 + 0 + 1^2 + 2^2 + 3^2 + 4^2 + 5^2}{12}$$

$$= \frac{1 + 4 + 9 + 16 + 25}{6} = \frac{55}{6} = 9.166$$

Population and sample:
A population is the entire group that you want to draw conclusions about.
A sample is the specific group that you collect data from.
The size of the sample is always less than the total size of the population.

sample variance $s^2 = \frac{\Sigma (x_i - \bar{x})^2}{n-1}$ as opposed to $\sigma^2 = \frac{\Sigma (x_i - \mu)^2}{N}$

s : S.D. of sample

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$$

Reason: This has bias if we use 'n' as we cannot estimate the s.D. of the population just from the sample.

Best case: Sample the data infinitely many times and then calculate mean, but we cannot do that, thus adding $\left(\frac{n}{n-1}\right)$ as the balancing factor for this bias.

eg. There are 45 students in a class. 5 students were randomly selected from the class and their heights were recorded as 131, 148, 139, 142, 152 cms.

$\bar{x} = 142.4$

$$s^2 = \frac{(-11.4)^2 + 5.6^2 + (-3.4)^2 + (-0.2)^2 + 9.6^2}{5-1} = 66.27$$

Variance for grouped data:

$$\sigma^2 = \frac{\sum f_i (x_i - \mu)^2}{\sum f_i} \qquad s^2 = \frac{\sum f_i (x_i - \mu)^2}{\sum f_i - 1}$$

eg:

Range	Frequency	x_i	$x_i - \mu$	$(x_i - \mu)^2$	$f_i(x_i - \mu)^2$
1-10	2	5.5	-17.39	302.41	604.82
11-20	7	15.5	-7.39	54.61	382.28
21-30	10	25.5	2.61	6.81	68.12
31-40	3	35.5	12.61	159.01	477.04
41-50	1	45.5	22.61	511.21	511.21

$$s^2 = \frac{\sum f_i (x_i - \mu)^2}{\sum f_i - 1} = 92.89$$

$$\sigma^2 = \frac{\sum f_i (x_i - \mu)^2}{N}$$

$$= \frac{1}{N} \sum f_i (x_i^2 - 2\mu x_i + \mu^2)$$

$$= \frac{\sum f_i x_i^2}{N} - 2\mu \frac{\sum f_i x_i}{N} + \mu^2 \frac{\sum f_i}{N}$$

$$= \frac{\sum f_i x_i^2}{N} - 2\mu^2 + \mu^2$$

$$\boxed{\sigma^2 = \frac{\sum f_i x_i^2}{N} - \mu^2} \quad = \frac{\sum f_i x_i^2}{N} - \left(\frac{\sum f_i x_i}{N}\right)^2$$

Coefficient of variation (CV)

It represents the ratio of standard deviation to the mean.

$$CV = \frac{\sigma}{\mu} \quad \text{or} \quad CV = \frac{s}{\bar{x}}$$

(often given as a percentage)

Q. CVs of 2 distributions are 55% and 65%, their SDs are 22 and 39 respectively. Find their AMs.

→

$$CV = \frac{\sigma}{\mu} \times 100 \qquad CV = \frac{\sigma}{\mu} \times 100$$

$$55 = \frac{22}{\mu} \times 100 \qquad 65 = \frac{39}{\mu} \times 100$$

$$\mu = 40 \qquad \mu = 60$$

Measures of association

Covariance:

It signifies the direction of the linear relationship between 2 variables.
By direction, we mean if the variables are directly or inversely proportional to one another.

$$cov(x,y) = \sum_{i=1}^{n} \frac{(x_i - \mu_x)(y_i - \mu_y)}{N} \quad \text{(population)}$$

$$cov(x,y) = \sum_{i=1}^{n} \frac{(x_i - \bar{x})(y_i - \bar{y})}{N-1} \quad \text{(sample)}$$

e.g.

	x	y
A	10	40
B	12	48
C	14	56
D	8	32
	$\mu_x = 11$	$\mu_y = 44$

$x - \mu_x$	$y - \mu_y$	$(x-\mu_x)(y-\mu_y)$
-1	-4	4
1	4	4
3	12	36
-3	-12	36
		80

$$cov(x,y) = \frac{80}{4} = 20 \Rightarrow \text{+ve covariance}$$

Correlation:

It tells how strong the relationship between 2 variables is.
It is simply covariance normalized.
-1 <= correlation <= 1

$$\text{correlation} = \frac{cov(x,y)}{\sigma_x \sigma_y}$$

$$= \frac{\sum \frac{(x_i - \bar{x})(y_i - \bar{y})}{N}}{\sqrt{\frac{\sum (x_i - \bar{x})^2}{N}} \sqrt{\frac{\sum (y_i - \bar{y})^2}{N}}}$$

Pearson's 1st coefficient of correlation

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \cdot \sum (y_i - \bar{y})^2}}$$

e.g.

	x	y
A	10	40
B	12	48
C	14	56
D	8	32
	$\mu_x = 11$	$\mu_y = 44$

$$\sigma_x = \sqrt{\frac{1+1+9+9}{4}} = \sqrt{5}$$

$$\sigma_y = \sqrt{\frac{16+16+144+144}{4}} = \sqrt{80}$$

$$r(x,y) = \frac{cov(x,y)}{\sigma_x \sigma_y}$$

$$= \frac{20}{\sqrt{5}\sqrt{80}} = 1$$

Summary

measures of central tendency

	ungrouped data	grouped data	grouped data (ranged)
mean (Am)	$\frac{x_1 + x_2 + \cdots + x_n}{n}$	$\frac{\sum f_i x_i}{\sum f_i}$	$\frac{\sum f_i x_i}{\sum f_i}$ ← x_i is midpoint of range
median	$\frac{n+1}{2}$ th term (odd); avg($\frac{n}{2}$ th, $\frac{n+1}{2}$ th) terms (even) ← Use cum. freq. to find these		$l + \frac{\frac{N}{2} - F}{f} \times h$
mode	value with highest freq.	value with highest freq.	$L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$

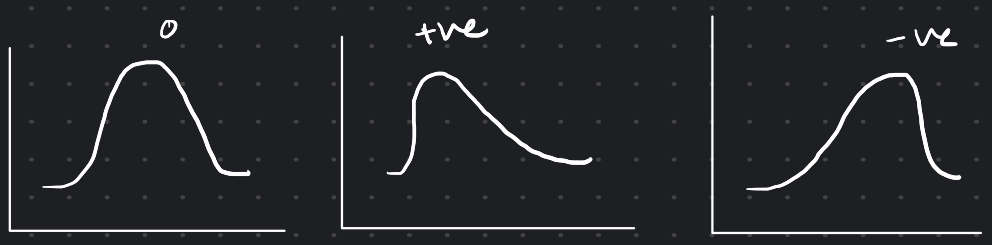
$$GM = \sqrt[n]{x_1 \cdot x_2 \cdot \cdots \cdot x_n} \qquad HM = \frac{n}{\frac{1}{x_1} + \cdots + \frac{1}{x_n}}$$

$$x_{RMS} = \sqrt{\frac{x_1^2 + x_2^2 + \cdots + x_n^2}{n}}$$

Empirical relation: $(\text{mean} - \text{mode}) \approx 3\,(\text{mean} - \text{median})$

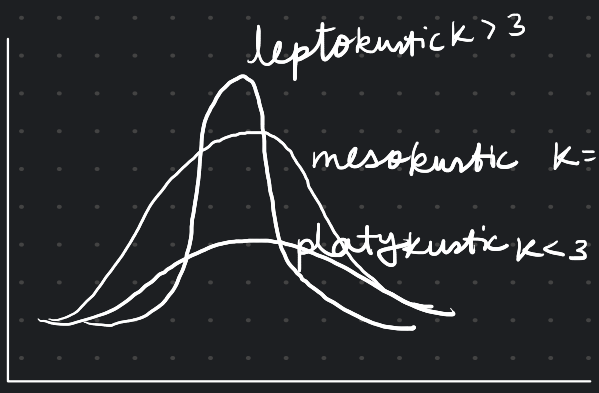
measures of shape:

Skewness



$$1^{st} \text{ coeff. of skewness} = \frac{\text{mean} - \text{mode}}{\text{std. devia}^n}$$

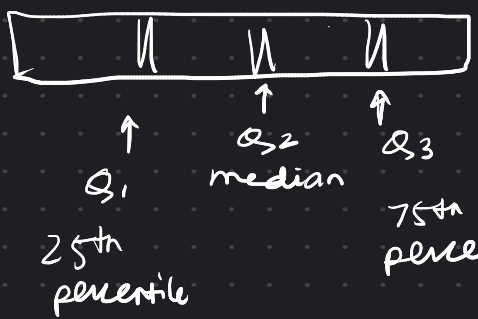
Kurtosis



measures of spread:

Range = highest value − lowest value

Quartiles:



$IQR = Q_3 - Q_1$

Variance:

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N} \qquad s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

Grouped data:

$$\sigma^2 = \frac{\sum f_i (x_i - \mu)^2}{N} \qquad s^2 = \frac{\sum f_i (x_i - \bar{x})^2}{\sum f_i - 1}$$

$x_i \rightarrow$ midpoint of range

coeff. of variation $= \frac{\text{std. deviation}}{\text{mean}}$

measures of association:

covariance:

$$cov(x,y) = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{N} \qquad cov(x,y) = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

correlation: $$r = \frac{cov(x,y)}{\sigma_x \sigma_y} = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\Sigma (x_i - \bar{x})^2} \sqrt{\Sigma (y_i - \bar{y})^2}}$$

Questions

Q. Std deviation of $\{18, 23, 14, 3, 17\}$ is

→ $\mu = \frac{75}{5} = 15$

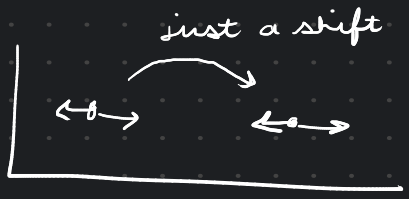
$$\sigma^2 = \frac{9 + 64 + 1 + 144 + 4}{5} = \frac{222}{5} = 44.4$$

$$\sigma = 6.66$$

Q. mean and std. dev^n of 75 observa's are 45 and 10 resp.
2 is added to each observa^n. mean & std. dev^n?

→ $\frac{\sum x_i}{N} = 45$ $\quad \sum x_i = 45 \times 75 = 3375$

$$\sum x_i' = 3375 + 2(75)$$



$$\mu' = \frac{3375 + 2(75)}{75} = \mu + 2 = 47$$

mean increased by 2.
Std devia^n remains the same.

$(\mu', \sigma') = (47, 10)$.

Q. Mean of 4 #s is 37, mean of smallest 3 is 34.
If range is 15, what is mean of largest 3?

→ $\frac{x_1 + x_2 + x_3 + x_4}{4} = 37$ $\quad x_1 + x_2 + x_3 = 34.3 = 102$

$$\frac{102 + x_4}{4} = 37$$

$$x_4 = 46$$

$$46 - x_1 = 15$$

$$x_1 = 31$$

$$\frac{31 + x_2 + x_3 + x_4}{4} = 37$$

$$x_2 + x_3 + x_4 = 117$$

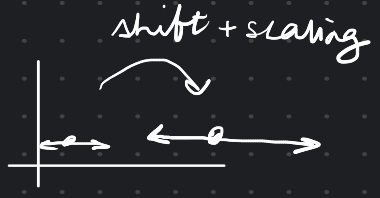
$$\frac{x_2 + x_3 + x_4}{3} = 39$$

Q. mean = 3, variance = 2 for 10 observa^ns. If each observa^n is multiplied by 2, new mean & variance?

→ $\frac{\Sigma x}{10} = 3$ $\quad \Sigma x = 30$

$\Sigma x' = 30 \cdot 2 = 60$

$\mu' = \frac{60}{10} = 6$



$$\sigma^2 = \frac{(x_1-\mu)^2 + \cdots + (x_n-\mu)^2}{n}$$

$$\sigma^{2'} = \frac{4(x_1-\mu)^2 + \cdots + 4(x_n-\mu)^2}{n}$$

$$= 4\sigma^2 = 4 \cdot 2 = 8$$

Q. $x_1\ x_2\ x_3\ x_4$ mean = 3, std. dev^n = 2.

mean & std. dev of $3x_1+5,\ 3x_2+5,\ 3x_3+5,\ 3x_4+5$ = ?

→ $\mu' = \frac{3(x_1+x_2+x_3+x_4) + 5(4)}{4} = 3 \cdot \mu + 5 = 14$

$\sigma^2 = 4$ Becomes 3^2 times i.e. = 36

$\sigma^{2'} = 36$

$\sigma' = 6$